Robust Multiview Point Cloud Registration Using Algebraic Connectivity and Spatial Compatibility

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***Abstract*—In this article, a spectral method for multiview point cloud registration is presented. The primary challenge of existing multiview registration methods is to accurately estimate the weights of the pose graph. Previous multiview registration methods rely on predicting the overlap between pairs of point clouds to assign weights for pairwise transformation matrices. This global way may result in higher weights being assigned to pairs of point clouds that are structurally similar but actually have no or low overlap. In addition, the learning-based approach is biased for training scenarios. However, we obtain the weights by measuring the confidence of each correspondence used for the estimation of the pairwise transformation matrix. Specifically, we construct spatial compatibility matrices for the correspondences and compute the edge weights of the pose graphs using spectral decomposition (SD). Considering the compatibility between correspondences, the proposed method is robust to outliers. Simultaneously, to mitigate the disruption of outliers to the global pose estimation and enhance the computational efficiency, we propose an overlap score estimation method based on the algebraic connectivity of the graph for pruning of fully connected pose graphs. For the initial sparse pose graph, we apply iteratively reweighted least-squares (IRLS) to refine the global transformation and design a new reweighting function based on the historical weighted average (HWA). The nonlearning framework empowers the proposed method to have better generalization to unknown scenarios. The experimental results further validate the performance of our method which achieves about 12.9% and 21.4% lower rotation and translation errors on the ScanNet dataset compared to the state-of-the-art method. The source code is available at https://github.com/swccj/SMVR.**

***Index Terms*—Algebraic connectivity, historical weighted average (HWA), iteratively reweighted least square (IRLS), multiview registration, spectral decomposition (SD).**

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I. Introduction

POINT cloud registration plays a critical role in numerous computer vision applications, including robotics [1], simultaneous localization and mapping (SLAM) [2], autonomous driving [3] and medical imagine [4]. In recent years, the field of pairwise registration has been fully developed, and many related works have sprung up. However, a completed scene may include several viewpoints in practical engineering where pairwise point cloud registration can only reconstruct part of the scene. Out of this need, multiview registration has been proposed and gained attention. By multiview point cloud registration, all the individual point cloud fragments in a scene are globally aligned to obtain a completed scene reconstruction. However, compared to pairwise registration, which is well studied, multiview registration works are still relatively scarce [5], [6], [7], [8], [9], [10].

Existing multiview registration methods generally apply a two-stage strategy to find the final globally consistent solution. The first step is to construct a sparse graph where the vertices represent the scans and the edges represent the relative transformation matrices between two scans. There are currently two options for constructing sparse graphs. One is to first construct a fully connected graph of all scans and then prune the edges based on the overlap between point cloud pairs [5], [6], [11]. The other is to add edges one by one until a connected graph is formed [12], [13]. A common practice is to construct the minimum spanning tree of the original fully connected graph [8]. Compared to a fully connected graph, a sparse graph is not only more robust to outliers but also reduces the subsequent computation and time consumption. The second stage is a global optimization based on graphs. The initial pose graph does not satisfy global consistency due to the inherent error in the estimated pairwise registration matrix and the accumulated errors. Most current methods prefer to use the iteratively reweighted least square (IRLS) mechanism to iteratively optimize and update the pose graph and global transformation matrices. The initial global transformation matrices are computed using the pose synchronization algorithm, followed by updating the edge weights of the pose graph based on the cycle consistency. Hence, the global transformation matrices and the pose graph are alternately updated.

Previous multiview point cloud registration methods generally rely on the prediction of point cloud overlap as the initial weights of the pose graph edges. These methods use the overlap rate as a measure of confidence for the pairwise transformation matrices. The state-of-the-art method [5] uses a neural network to extract the global feature of a point cloud and predicts the overlap rate by comparing the global features of two point clouds. However, it may not be reliable when encountering point cloud pairs that are similar in overall structure but do not actually overlap. Moreover, the bias of learning-based methods renders them not well generalizable to unknown scenarios. To address these issues, in this article, we propose a new perspective to evaluate the confidence weights of relative transformation matrices. Our method focuses on the confidence of each correspondence rather than the global feature that leads to the overall ambiguity problem mentioned above. This also explains the fact that our method is only applicable to the correspondence-based pairwise point cloud registration method.

The proposed method is based on the spectral technique [14], [15], [16], [17], [18], [19]. First, we construct the weighted adjacency matrix for all correspondences. Then, we use a spatial compatibility measure to calculate the compatibility of each correspondence with the others to assign value to the adjacency matrix. This adjacency matrix is called the spatial compatibility matrix. If two correspondences satisfy the spatial consistency (i.e., the distance between the two points remains the same before and after the rigid transformation), they will receive a higher compatibility score. We take a widely adopted assumption from [14] that inliers will form a strongly connected main cluster since they exhibit mutual compatibility under the constraint of spatial consistency. Outliers tend to be outside the main cluster or remain weakly connected to it due to the small probability that they can form compatible relationships with other correspondences. The connections between outliers are always random and unstructured, which results in their inability to form clusters that threaten the dominance of the cluster of inliers. Next, we perform a spectral decomposition (SD) on the obtained spatial compatibility matrix. From this, we can get the connection (i.e., strong or weak) between each correspondence with the main cluster and take the connection as the reliability of each correspondence being inliers. This connection is then used to compute an overall intercluster score which measures the quality of the correspondence set. Since the pairwise transformation matrix is estimated on the correspondence set, this score naturally represents the confidence of the relative pose. However, the construction of spatial compatibility matrices on the entire fully connected pose graph requires significant computing time, and the SD results of unreliable edges may impact the final weights. Therefore, we propose an overlap prediction approach based on the algebraic connectivity of the graph, along with a corresponding pruning scheme to be applied before constructing the spatial compatibility matrices. In the phase of global graph optimization, we design a new reweighting function based on the historical weighted average (HWA). The IRLS algorithm takes the pose graph obtained in the first step as an initial input and updates the weights and global transformation matrices in an alternating iterative manner. During the iteration process, inliers and outliers are assigned higher and lower weights, respectively, until convergence is reached. By taking the history weights into account, the reweighting function can have better robustness to outlier edges and eventually converge to the correct poses.

The proposed method fundamentally operates on a nonlearning framework. Unlike learning-based approaches, our method obviates the requirement for extensive datasets for training. Furthermore, the nonlearning paradigm inherently avoids bias toward training scenarios, thereby exhibiting superior generalization capabilities. We have evaluated the proposed method on three datasets: 3D(Lo)Match [20], [21], ScanNet [22], and ETH [23]. The state-of-the-art method [5] is trained in 3DMatch and tested on other datasets. The experimental results show that our method achieves about 0.5% higher registration recall (RR) on 3DMatch. However, this gap is further stretched on ScanNet with 12.9% and 21.4% lower rotation and translation errors, which means that our method has a better performance and avoids the bias of the learning-based approach.

Our main contributions are summarized as follows.

1) We introduce a sparse pose graph construction method that incorporates edge weight prediction based on SD with a novel pruning scheme to avoid the inherent bias of the learning-based approach for the training scenarios.
2) We propose a new reweighting function based on the HWA for IRLS that exploits the error from previous iterations for better robustness to outliers.
3) The proposed method is thoroughly evaluated on three commonly used datasets, further demonstrating the superior performance and generalization capabilities of our method.

II. Related Work

A. Pairwise Registration

Traditional methods for aligning pairwise point clouds include two main categories: feature-based methods and optimization-based methods. The feature-based methods first detect key points [21], [24] and then extract local features [24], [25], [26], [27], [28], [29] with rotation-invariant properties for each key point. Traditional hand-crafted local feature descriptors for point clouds rely on features such as normals, colors, and distances. Deep learning-based methods, on the other hand, learn more descriptive features by mapping low-dimensional feature vectors into high-dimensional feature spaces. Then, a nearest-neighbor search is performed in the mapped feature space to obtain the correspondence of each key point. Some methods take the option to conduct outliers removal before the final transformation matrix estimation to obtain more reliable results [16]. Optimization-based registration [30], [31] methods use an optimization strategy to estimate the transformation matrix. Most optimization-based methods consist of two stages: correspondence search and transformation estimation. These two phases will be alternately iterated to find the best transformation. During the iteration process, the initial correspondence may not be accurate. As iterations continue, the correspondence will become more accurate. The estimated transformation matrix is then accurate by using the exact correspondence. Classical methods in this category include ICP-based methods [32], [33], [34], graph-based methods [14], [35], [36], and probabilistic-based methods [37], [38]. In addition to traditional alignment methods, some end-to-endpoint cloud registration networks [15] based on deep learning have recently appeared.

B. Multiview Registration

Early multiview point cloud registration is achieved by successive pairwise registration. This practice is obviously extremely sensitive to accumulated errors, which can ultimately lead to results that are far from expected. The modern method [5], [6], [7], [8], [39], [40] first constructs a graph of relative poses in which nodes represent point clouds and edges represent relative poses between two point clouds. Then, the global poses are iteratively optimized according to the cycle consistency, and finally, the robust global poses are obtained. The input to the optimization process is a weighted graph, where the weight of each edge represents the confidence of the corresponding relative transformation matrix. Recent methods commonly use the predicted overlap between a pair of point clouds as an indirect confidence of the relative pose and set it as the edge weight. Wang et al. [5] used a neural network to extract the global features of the point cloud. Then, they computed the weight of the edge by comparing the global feature similarity of two point clouds and combining it with the inlier rate. Gojcic et al. [6] proposed an end-to-end network to predict the weight and final pose. Zhao et al. [8] considered overlap points to have similar distances in the feature space and used this to calculate the overlap confidence. However, our method derives the confidence of the relative pose by evaluating the quality of each correspondence.

When constructing the pose graph, a sparse graph not only saves a lot of computational consumption but also removes the effect of outliers to some extent. The current sparse pose graph construction methods mainly include two kinds: one is to construct the fully connected graph first, and then prune it based on overlap confidence [5], [6], [41], [42], [43]; the other is to construct the minimum spanning tree [8], [11], [13]. In the final phase of global graph optimization, the IRLS algorithm is widely adopted [6], [7], [41], [44], [45], [46] and the key point is the reweighting function. Data-driven learning-based methods [6], [7], [47] often have good performance in weight updating, but the generalization of this approach is yet to be improved. Wang et al. [5] proposed a new history reweighting function that has strong generalization while maintaining excellent robustness. Motivated by [5], we design a new historical weighted average function to reweight the weight during the iteration. It shows good robustness to outliers.

III. Method

A. Overview

Consider a point cloud set $S = \{S_i | i = 1, \dots, N_S, S_i \in \mathbb{R}^{N\times 3}\}$, where each scan corresponds to a viewpoint in a complete scene. The object of multiview point cloud registration is to compute the rigid transformation $T = \{T_i = (R_i, t_i) \in \mathrm{SE}(3) | i = 1, \dots, N_s\}$ of the scans to the global coordinate system. In this section, we will derive the specific steps of the proposed multiview registration method. First, we will show how to predict the overlap and confidence of the transformation matrix for pairwise point cloud registration and initialize the sparse pose graph based on the calculated confidence in Section III-B. In Section III-C, we describe how to use the idea of HWA in combination with the IRLS algorithm to iteratively optimize the pose graph for a final reliable solution. The overview of the proposed method is illustrated in Fig. 1.

B. Sparse Graph Construction

In this section, we exhibit how to evaluate the overlap score and the confidence of the pairwise registration matrix for each pair scan, followed by the construction of a sparse pose graph among the scans. For each pair of scans (S_i, S_j), we first extract their correspondence set C_{ij}. The overlap score is derived from the algebraic connectivity of the feature connection graph for C_{ij}. Then, we assign a vector x, where $x \in \mathbb{R}^{N_c\times 1}$ and N_c is the number of correspondences in C_{ij}, to indicate the confidence of each individual correspondence by SD of the spatial compatibility matrix M. In the following, we use the confidence vector x to compute the score s_{ij} of the whole correspondence set as the confidence of the relative pose between the two point cloud scans. Finally, we construct the fully connected graph, then prune and initialize it based on the overlap score and the relative pose confidence.

1) Correspondence Extraction: To build the correspondence set, we first extract the local feature $f_i \in \mathbb{R}^d$ of each point in point cloud S_i. We take YOHO [28] as the local feature descriptor. The raw point cloud is voxelized and downsampled, and then the YOHO feature is computed for each point. For point clouds S_i and S_j, we search for the nearest neighbor in the feature space as the corresponding point for each point in S_i. The correspondence (i, j) is built as follows:

$$j = \arg\min_j \| f_i - f_j \| \tag{1}$$

where $(f_i, f_j) = \mathrm{YOHO}(i, j), i \in S_i, j \in S_j$ and $\|\cdot\|$ represents the $L2$-norm. Finally, the correspondence C_{ij} between the two point clouds is established.

2) Prediction for the Overlap Score: Directly constructing spatial compatibility matrices on the fully connected pose graph not only costs considerable computational time but is also prone to the impact of outlier edges. Hence, before estimating the relative pose confidence, we first apply a pruning scheme based on the algebraic connectivity of the graph. To predict the overlap score for each pair of scans (S_i, S_j), we consider the points of their correspondence set C_{ij} as potential overlap points. Then, we determine the overlap score by associating the algebraic connectivity of the graph, formed by the potential overlap point features, with the actual overlap regions.

The main idea of this approach can be explained with a simple illustration in Fig. 2. Previous studies revealed that correspondences in low overlap point clouds tend to fall into nonoverlapping regions, resulting in worse alignment performance [21]. In cases of high overlap point clouds, the majority of potential overlap points lie within the same

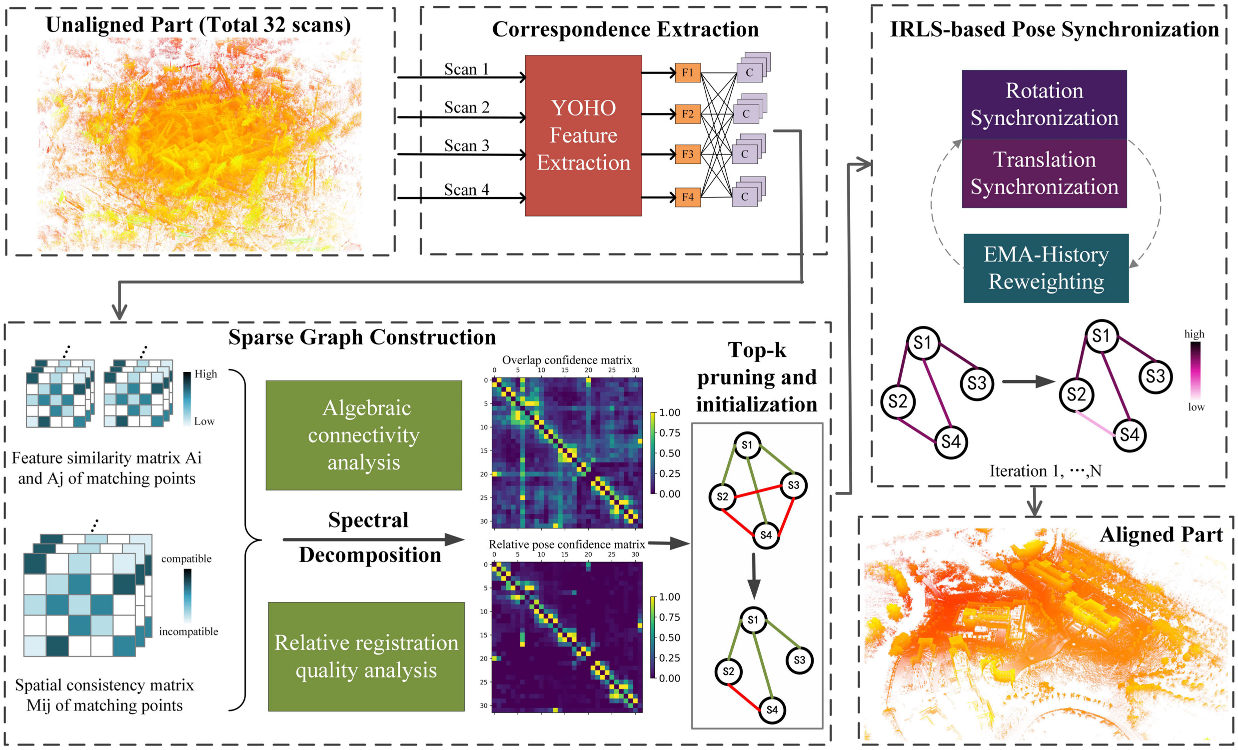

Fig. 1. Overview of the proposed method. Given a point cloud set, the correspondences C are first extracted, then the overlap score is computed and the pose graph is pruned according to the algebraic connectivity of graphs. Subsequently, the relative pose confidence is obtained as the pose graph weights by SD of the spatial compatibility matrix. Finally, the proposed IRLS scheme is utilized for the iterative refinement of the global pose.

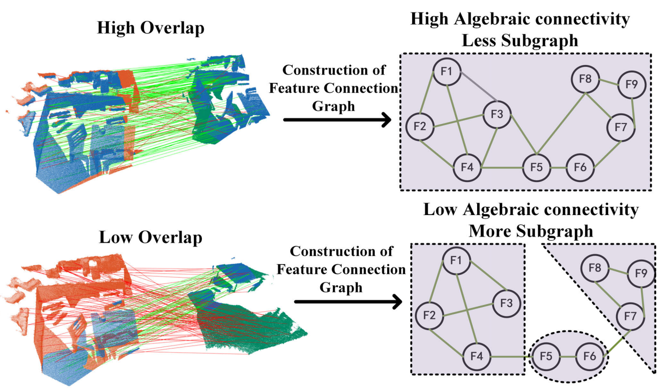

Fig. 2. Relationship between the overlap ratio and the algebraic connectivity of feature connection graph. The overlap region, correct correspondence, and incorrect correspondence between the source point cloud and the target point cloud are indicated by blue points, green lines, and red lines, respectively.

overlap region, leading to features that exhibit strong similarity and correlation. Consequently, the graph constructed based on feature similarity demonstrates greater connectivity and encompasses fewer subgraphs. For the low overlap point clouds, the feature distributions of the potential overlap points are more diverse and distinct due to falling in different regions. Hence, the connected graphs exhibit reduced connectivity and contain more subgraphs.

Specifically, we construct a connected graph G_i^c and G_j^c of the potential overlapping point features for a pair of scans (S_i, S_j), respectively, and then approximate the overlap score of (S_i, S_j) via the sum of algebraic connectivity of the graph G_i^c and G_j^c. The adjacency matrix A_i of the connected graph G_i^c is defined as

$$A_i = F_i F_i^{\mathrm{T}} \tag{2}$$

where $F_i \in \mathbb{R}^{N_c \times d}$ is the feature matrix of the matching points in scan S_i and d is the dimension of YOHO descriptor. Hence, we obtain a connected graph G_i^c of matching point features with N_c nodes, where the edges represent the similarity of features. The Laplacian matrix Q_i of graph G_i^c is an $N_c \times N_c$ matrix, which is defined as

$$Q_i = \triangle - Ai \tag{3}$$

where $\triangle = \mathrm{diag}(D_m)$. D_m is the nodal degree of the node $m \in N_c$ and diag denotes the diagonal matrix with nodal degree D_m as diagonal elements. The eigenvalues of Q_i are called the Laplacian eigenvalues, that is, $\lambda_{N_c} = 0 \leq \lambda_{N_c-1} \leq, \ldots, \leq \lambda_1$. The second smallest Laplacian eigenvalue λ_{N_c-1}, also known as algebraic connectivity, was extensively studied [48], [49], [50]. The most important is that the larger the algebraic connectivity is, the more difficult it is to cut a graph into independent components. Here, it is assumed that due to the high overlap ratio, most of the matching points tend to fall into the same overlap region, resulting in their features exhibiting significant similarity and correlation. Therefore, the overlap score of the scans (S_i, S_j) is defined as the sum of λ_{N_c-1} of graph G_i^c and G_j^c, and normalized to the range [0, 1].

Following [5], we adopt the top-k algorithm as a pruning strategy to construct a sparse graph. For each scan, we select scan pairs of the k highest overlap score to build connections. This reduces the computational cost and the interference from outlier edges subsequently. The weights of the connections are initialized by the subsequent relative pose confidence.

3) Prediction for Relative Pose Confidence: Given two pairs of correspondence (i, j) and (i', j') in C_{ij}, they are supposed to satisfy the spatial consistency under rigid transformation if they are inliers, which is called spatial compatibility. The spatial compatibility between (i, j) and (i', j') is calculated as follows [15]:

$$\mathrm{SC} = \left[1 - \frac{d_{ij}^2}{\theta_d^2}\right]_+ \tag{4}$$

$$d_{ij} = \left|\left\|i - i'\right\| - \left\|j - j'\right\|\right| \tag{5}$$

where $[\cdot]_+$ is the $\max(\cdot, 0)$ operation and θ_d represents the dividing value for compatibility between two correspondences. The SC score is a well-defined measure of spatial compatibility and relaxes this compatibility to [0, 1].

For the correspondence C_{ij}, we construct its adjacency matrix $M \in \mathbb{R}^{N_c \times N_c}$, where N_c is the number of correspondences in C_{ij} and $M_{ii'}$ denotes the SC score between correspondence (i, j) and (i', j') in C_{ij}. The adjacency matrix reflects the spatial compatibility between all correspondences and is defined as the spatial compatibility matrix. From the definition of spatial compatibility, we can tell that M is a sparse nonnegative and symmetric matrix.

Following [14], [15], and [16], we argue that inliers will constitute a strongly connected cluster under spatial consistency constraints, whereas outliers do not tend to share this property due to their small probability of being highly spatially compatible with other correspondences and the random unstructured way in which they form these compatibility relationships. We use an intercluster score [14] to describe the reliability of the whole correspondence. The intercluster score is defined as

$$s = x^{\mathrm{T}} M x \tag{6}$$

where $x \in \mathbb{R}^{N_c \times 1}$ and x is interpreted as the association between each correspondence and the main cluster. Thus, finding the main cluster is equivalent to solving for a x that maximizes the intercluster score s. Rayleigh's theorem proves that x, which maximizes the intercluster score $x^{\mathrm{T}} M x$, is the principal eigenvector of matrix M. We use the power iteration algorithm to get the principal eigenvector. According to the Perron–Frobenius theorem, the elements of x will be in [0, 1] because M is a nonnegative matrix. Since the main cluster is composed of inliers, we can naturally reinterpret x as the reliability of each correspondence being an inlier. After calculating the intercluster score s for all scan pairs, we normalized s to serve as the final confidence of the relative transformation matrix estimated based on the whole correspondence

$$s_{ij} = \frac{s_{ij}}{\mathrm{norm}(S)} \tag{7}$$

where i and j represent the ith and jth scans, respectively. S is the collection of the intercluster score of all scan pairs. $\mathrm{norm}(\cdot)$ denotes the $L2$-norm.

For the scan pairs establishing connections based on overlap scores, we utilize intercluster scores to initialize the edge weight as relative pose confidence and then utilize RANSAC to estimate the relative transformation matrix on the extracted correspondence. This approach avoids having to compute the pairwise transformation matrices between all scans but also reduces the impact of outlier edges and saves a lot of subsequent computational consumption.

C. IRLS-Based Pose Optimization

In this section, we present how the IRLS algorithm works for the solution of the global transformation matrix for each point cloud. First, the intercluster score that was earlier computed to measure the confidence of the relative pose is now used to assign initialization weight to each edge of the sparse graph. Then, we use the weights as input to the pose synchronization algorithm to correct the global poses obtained in the last step. In the following, we update the weights based on cycle consistency to make the weights of the inlier edge larger and the weights of the outlier edge smaller. The synchronization of the transformation and the refinement of the weights are conducted alternately and iteratively to obtain the final solution with global consistency.

Following [5] and [6], we adopt the option to decompose the transformation synchronization problem into rotation synchronization [7], [51] and translation synchronization [6], [51], due to their guaranteed closed-form solutions. Given the input relative transformation matrix $T_{ij} = \{R_{ij}, t_{ij}\}$ and its initial weights w_{ij}, the objective of the transformation synchronization algorithm is to derive a synchronized global pose $T_i = \{R_i, t_i\}$ for each point cloud. Here, we use the earlier calculated intercluster scores S as the initial weights.

1) Rotation Synchronization: Solving global rotation matrices by rotation synchronization algorithm based on relative transformation matrices and their weights can be transformed into finding the solution of an optimization problem

$$R_i = \arg\min_{R_i} \sum_{(i,j)\in\varepsilon} w_{ij} \left\| R_{ij} - R_i R_j^{\mathrm{T}} \right\|_F^2 \tag{8}$$

where ε represents the edge set of the sparse pose graph and $\|\cdot\|_F$ denotes the operation of computing the Frobenius norm of the matrix. We adopt the algorithm in [7] to find a closed-formed solution. The algorithm starts by building the Laplace matrix $L \in \mathbb{R}^{3n \times 3n}$ of the pose graph. Naturally, L is a sparse symmetric matrix due to the sparsity of the pose graph. L is defined as

$$L_{ij} := \begin{cases} \sum_{j\in N(i)} w_{ij} I_3, & i = j \\ -w_{ij} R_{ij}^{\mathrm{T}}, & (i, j) \in \varepsilon \\ 0, & \text{otherwise} \end{cases} \tag{9}$$

where $N(i)$ denotes the nodes that are connected to i in pose graph. Then, the matrix L is spectrally decomposed and the eigenvectors corresponding to the three smallest eigenvalues are taken and used to construct the matrix $U = (U_1^{\mathrm{T}}, \ldots, U_n^{\mathrm{T}})^{\mathrm{T}} \in \mathbb{R}^{3n \times 3}$. Finally, the corresponding absolute rotation is solved by performing singular value decomposition (SVD) on each U_i.

2) Translation Synchronization: Similar to rotation synchronization, solving translation vector t_i is also a least-squares problem

$$R_i = \arg\min_{R_i} \sum_{(i,j)\in\varepsilon} w_{ij} \left\| R_{ij} t_i + t_{ij} - t_j \right\|^2. \tag{10}$$

The process of solving for the translation vector t_i is provided by [5]. First, matrices $A \in \mathbb{R}^{3E*3E}$ and $B \in \mathbb{R}^{3E*3N}$ are initialized as identity and zero matrices, respectively, where E represents the number of connected edges in pose graph. In addition, a vector $H \in \mathbb{R}_{3E*1}$ is constructed. For the eth edge (i, j), $A[3e-3{:}3e]$ is multiplied by w_{ij} and the (e, i) and (e, j) $3*3$ blocks of B are filled with $-I_3$ and I_3, respectively. Then, $R_i t_{ij}$ is filled to the eth $3*3$ block of H. Finally, the translation vector t_i of each scan P_i is obtained by solving $t = (B^{\mathrm{T}} AB)^{-1} B^{\mathrm{T}} AH$.

3) Weight Updating Based on HWA: In each iteration of the refinement, we want the weights of the outlier and inlier edges to decrease and increase stably, respectively. To achieve a smooth and reliable convergence of the weights, we introduce the idea of HWA to design a new weight update strategy.

For the global pose input $\{T_i^{(n)} = (R_i^{(n)}, t_i^{(n)})\}$ in the nth iteration, we first compute the rotational residual $\delta_{ij}^{(n)}$ [5]. The rotational residual is calculated as follows:

$$\delta_{ij}^{(n)} = \triangle\left(R_{ij}, R_i^{(n)T} R_j^{(n)}\right) \tag{11}$$

where $\triangle(\cdot)$ represents the angular between two rotation matrices. Then, we design a new error calculation based on exponential moving average (EMA). The error consists of two parts and is calculated as follows:

$$v_{ij}^{(n+1)} = \theta_1 + \theta_2 \tag{12}$$

$$\theta_1 = \lambda_\alpha \left(\delta_{ij}^{(n+1)} - \delta_{ij}^{(n)}\right) \tag{13}$$

$$\theta_2 = (1 - \lambda_\alpha - \lambda_\beta) v_{ij}^{(n)} + \lambda_\beta \delta_{ij}^{(n+1)} \tag{14}$$

where θ_1 describes the change in the rotational residuals over two iterations. This change can guide the reweighting function to update the weights in the direction of error reduction in each iteration, and as it approaches close to zero, the weights are close to convergence. θ_2 represents the exponential moving average of the rotated residuals over recent iterations. With the help of EMA, we can obtain a relatively smooth curve of error changes, thus minimizing the influence of outliers.

In terms of the weight update, we use an HWA to predict the weight of the current iteration round. First, we use the error $V_{ij} = \{v_{ij}^{(m)} | m = 1, \ldots, n\}$, where m denotes the mth iteration, to compute the historical weight vector $g \in \mathbb{R}^{n\times 1}$ of the edge weight. The historical weight vector G is computed by

$$g(m) = \frac{e^{-v_{ij}^{(m)}}}{\left\| \left(e^{-v_{ij}^{(1)}}, \ldots, e^{-v_{ij}^{(n)}}\right) \right\|_2} \tag{15}$$

where $\|\cdot\|_2$ denotes the $L2$-norm. By comparing the relative size of the errors across all iteration rounds, we compute a history weight for the edge weights in each history iteration. Finally, we obtain the edge weights in the current iteration by computing the HWA of the edge weights in all iterations. Therefore, the edge weight in the $(n+1)$th iteration can be calculated by

$$w_{ij}^{(n+1)} = \sum_{m=1}^{n} g(m) w_{ij}^{(m)}. \tag{16}$$

Although the calculation of the proposed EMA-based error can obtain a smoother and more reliable error carve, the top-k-based pruning method still retains a large number of outlier edges and these outliers still disturb the updating of weights considerably. In this case, the reweighting function based on HWA is more robust to outliers. It ensures a satisfactory convergence of the weights and a numerical difference between the inliers and the outliers.

After the weights have been computed, the global poses and weights are updated alternately and iteratively to get a final stable solution with global consistency.

IV. Experiments

A. Datasets and Experimental Setup

We evaluate our method on four datasets: 3D(Lo)Match [20], [21], ScanNet [22], ETH [23], and WHU-TLS [52].

1) 3D(Lo)Match: 3DMatch is an RGB-D dataset consisting of 62 real indoor scenes where 54 scenes are split for the training set and the remaining 8 are used as the test set. The separate scenes of 3DMatch are divided into a number of point cloud scans, where each test scene contains an average of 54 scans. The official 3DMatch dataset only considers scan pairs with >30% overlap, which is currently often used. Huang et al. [21] proposed a 3DLoMatch dataset, which only considers point cloud pairs with 10%–30% overlap. Compared to 3DMatch, 3DLoMatch is a more challenging dataset and we apply our method both on 3DMatch and 3DLoMatch to evaluate the performance at low overlap.

2) ScanNet: ScanNet is a large RGB-D video dataset with a total of 2.5 million views in 1513 indoor scenes. It is captured by an RGB-D capture system and annotated with 3-D camera poses, surface reconstructions, and instance-level semantic segmentations. For a fair comparison, we adopt the same test set as in [5] and [6]. The test set contains 32 randomly sampled scenes and 30 RGB-D images that are 20 frames apart are randomly sampled in each scene. These RGB-D images are converted to a point cloud dataset containing 32 scenes each including 30 scans. All the scan pairs are used for evaluation.

3) ETH: ETH is an outdoor point cloud dataset containing eight point cloud sequences that are acquired in locations covering the environment diversity that modern robots are susceptible to encounter, ranging from inside an apartment to a woodland area. We follow [5] to use the same four outdoor scenes with 713 pairs of scans for evaluation.

4) WHU-TLS: The WHU-TLS dataset is a large-scale benchmark dataset that encompasses 11 distinct environments, such as metro stations, high-speed rail stations, mountains, forests, parks, campuses, residential areas, river banks, cultural heritage buildings, subterranean mines, and tunnels. In this study, the WHU-TLS park dataset was employed for experimental analyses. This outdoor dataset comprises 32 scans and encompasses 160.24 million data points. It serves as

TABLE I
QUANTITATIVE COMPARISON ON THE SCANNET DATASET. EXCEPT FOR LMVR [6], WHICH INCLUDES PAIRWISE REGISTRATION IN ITS METHOD, ALL METHODS UTILIZE YOHO [28] AS PAIRWISE REGISTRATION ALGORITHM

Pose graph	Method	Rotation Error						Translation Error(m)					
		3°	5°	10°	30°	45°	Mean/Med	0.05	0.1	0.25	0.5	0.75	Mean/Med
Full	EIGSE3 [41]	19.7	24.4	32.3	49.3	56.9	53.6°/48.0°	11.2	19.7	30.5	45.7	56.7	1.03/0.94
	L1-IRLS [53]	38.1	44.2	48.8	55.7	56.5	53.9°/47.1°	18.5	30.4	40.7	47.8	54.4	1.14/1.07
	RotAvg [53]	44.1	49.8	52.8	56.5	57.3	53.1°/44.0°	28.2	40.8	48.6	51.9	56.1	1.13/1.05
	LITS [47]	52.8	67.1	74.9	77.9	79.5	26.8°/27.9°	29.4	51.1	68.9	75.0	77.0	0.68/0.66
	HARA [54]	54.9	64.3	71.3	74.1	74.2	32.1°/29.2°	35.8	54.4	66.3	69.7	72.9	0.87/0.75
	LMVR [6]	48.3	53.6	58.9	63.2	64.0	48.1°/33.7°	34.5	49.1	58.5	61.6	63.9	0.83/0.55
	SGHR [5]	57.2	68.5	75.1	78.1	78.8	26.4°/19.5°	39.4	61.5	72.0	75.2	77.6	0.70/0.59
	Ours	59.0	71.3	79.9	84.0	85.1	20.1°/17.8°	39.6	62.7	75.9	79.9	82.1	0.56/0.47
Pruned	EIGSE3 [41]	40.8	46.3	51.9	61.2	65.7	40.6°/37.1°	23.9	38.5	51.0	59.3	66.1	0.88/0.84
	L1-IRLS [53]	46.3	54.2	61.6	64.3	66.8	41.8°/34.0°	24.1	38.5	48.3	55.6	60.9	1.05/1.01
	RotAvg [53]	50.2	60.1	65.3	66.8	68.8	38.5°/31.6°	31.8	49.0	58.8	63.3	65.6	0.96/0.83
	LITS [47]	54.3	69.4	75.6	78.5	80.3	24.9°/19.9°	31.4	54.4	72.3	76.7	79.6	0.65/0.56
	HARA [54]	55.7	63.7	69.0	70.8	72.1	34.7°/31.3°	35.2	53.6	65.4	68.6	71.7	0.86/0.71
	SGHR [5]	**59.4**	71.9	80.0	82.1	82.6	21.7°/19.1°	**39.9**	63.0	74.3	77.6	80.2	0.64/0.47
	SGHR [5]*	59.1	**73.1**	**80.8**	82.5	83.0	21.7°/19.0°	**39.9**	**64.1**	**76.7**	79.0	81.9	0.56/0.49
	Ours	59.2	71.2	80.2	84.1	85.1	19.8°/**17.5°**	39.5	63.1	76.0	79.9	82.4	0.55/0.45
Ours	Ours	58.0	71.4	80.0	**86.0**	**86.5**	**18.9°**/17.6°	39.0	61.7	76.6	**81.5**	**84.8**	**0.46/0.44**

* means using top$k$ pruning scheme in [5].

a representative environmental model, incorporating a blend of anthropogenic structures, and is deemed suitable for the application of large-scale multiview registration techniques.

When constructing the spatial compatibility matrix, we set θ_d as 0.15 for 3DMatch, 0.06 for ETH, and 0.01 for ScanNet. For pruning, we set k as 10 for 3DMatch and ScanNet, 15 for 3DLoMatch, and 8 for ETH. In pose graph optimization, both λ_α and λ_β are set to 0.01.

B. Metrics

RR is used to report the percentage of correctly registered point cloud pairs out of all point cloud pairs. Following [5], we consider a pair of point clouds to be successfully aligned when the average distance between the points after the estimated transformation T_{pre} and the ground-truth (GT) transformation T_{gt}, respectively, does not exceed the set threshold. For a fair comparison, we set the same threshold as [5], 0.2 m for 3D(Lo)Match and 0.5 m for ETH. When evaluated on ScanNet, we adopt empirical cumulative distribution functions (ECDFs) of the rotation error re and translation error te [5], [6], [7] as follows:

$$\text{re} = \arccos\left(\frac{\text{tr}(R_{\text{pre}}^{\text{T}} R_{\text{gt}}) - 1}{2}\right), \quad \text{te} = \left\|t_{\text{pre}} - t_{\text{gt}}\right\|^2. \tag{17}$$

C. Baseline Approaches

We compare the proposed methods with some previous baseline methods to verify the overall effectiveness. EIGSE3 [41] is a method based on spectral techniques to solve the problem of transform synchronization and using IRLS to update the weights. RotAvg [53] proposes a generalized framework of relative rotation averaging and adopts the Lie algebra as the reweighting function. LMVR [6] is an end-to-end registration algorithm, which includes an initial pairwise registration and global consistency refinement. SGHR [5] proposes a history reweighting function, which is robust to the outlier and shows state-of-the-art performance. In addition to this, we also compared our method with some other works: L1-IRLS [53], LITS [47], and HARA [54]. Following [5], we use RANSAC as the pairwise registration estimation method for all methods except LMVR. We also report on two types of pose graph construction for a more comprehensive comparison. "Full" means a fully connected graph without any pruning strategy. "Pruned" indicates that the sparse pose graph is pruned by the pruning scheme in [6] by default, except for the one with *, utilizing the top-k pruning scheme provided in [5].

D. Results on Qualitative and Quantitative Comparisons

Tables I–III illustrate the quantitative comparison results on ScanNet, 3D(Lo)Match, and ETH, respectively. The qualitative results for indoor and outdoor scenarios are also shown in Figs. 3 and 4. In addition, qualitative results on the WHU-TLS dataset [52] are provided, which is a large-scale point cloud dataset for outdoor scenes. Fig. 4 illustrates the registration results and details of the proposed method in the park scene of the WHU-TLS dataset. It is evident from the details of buildings and trees that all point clouds have been accurately registered globally. The quantitative results show that the proposed method achieves different degrees of improvement on all three indoor and outdoor datasets. Our method obtains 0.5% and 1% improvement in RR for 3DMatch and 3DLoMatch, respectively. Compared to the state-of-the-art method, our method still has an advantage in 3DMatch and 3DLoMatch although the state-of-the-art method is trained in 3DMatch and tested on all datasets. Benefiting from the good performance of the deep learning network, the state-of-the-art method shows a considerable accuracy improvement on 3DMacth and

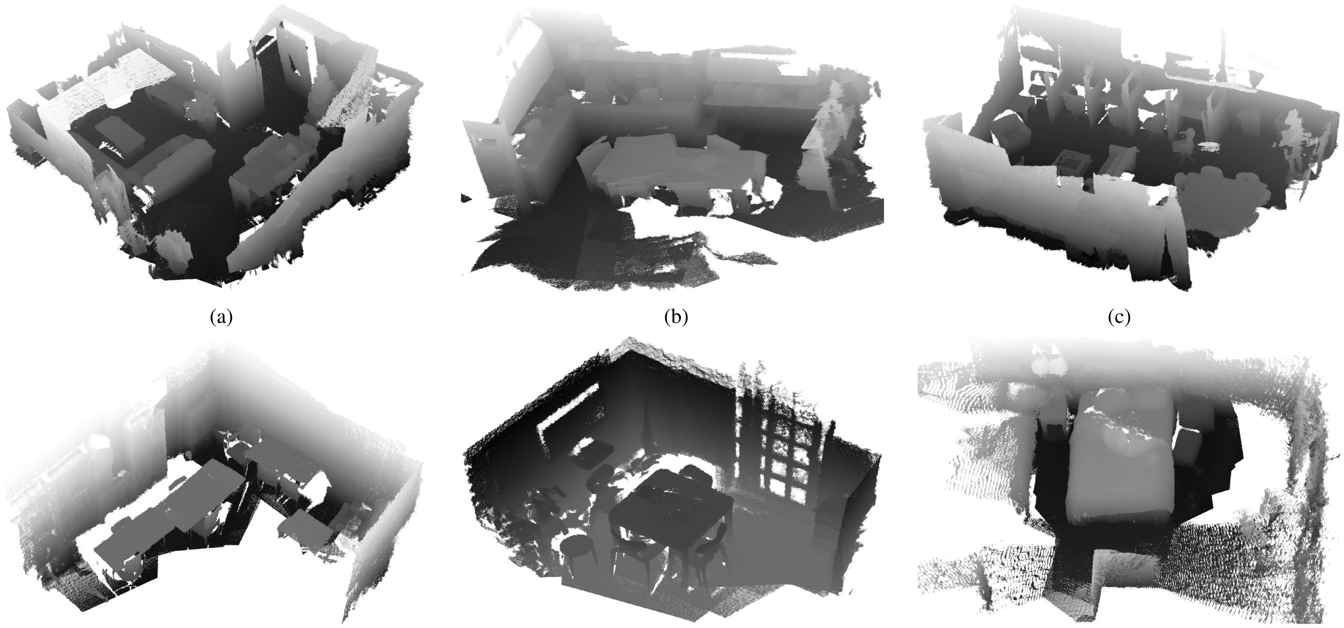

Fig. 3. Qualitative results on (top) 3DMatch and (bottom) ScanNet datasets, covering multiple indoor scenarios. (a) 3DMatch-home1 (60 scans). (b) 3DMatch-kitchen (60 scans). (c) 3DMatch-studyroom (66 scans). (d) ScanNet-0030-02 (30 scans). (e) ScanNet-0335-02 (30 scans). (f) ScanNet-0477-01 (30 scans).

TABLE II
QUANTITATIVE COMPARISON ON THE 3D(LO)MATCH. EXCEPT FOR LMVR [6], WHICH INCLUDES PAIRWISE REGISTRATION IN ITS METHOD, ALL METHODS UTILIZE YOHO [28] AS PAIRWISE REGISTRATION ALGORITHM

Pose graph	Method	3DMatch RR(%)	3DLoMatch RR(%)
Full	EIGSE3 [41]	23.2	6.6
	L1-IRLS [53]	52.2	32.2
	RotAvg [53]	61.8	44.1
	LITS [47]	77.0	59.0
	HARA [54]	83.1	68.7
	SGHR [5]	93.2	76.8
	Ours	93.8	77.6
Pruned	EIGSE3 [41]	40.1	26.5
	L1-IRLS [53]	68.6	49.0
	RotAvg [53]	77.2	60.3
	LITS [47]	80.8	65.2
	HARA [54]	83.8	71.9
	SGHR [5]*	95.2	82.3
	SGHR [5]	96.2	81.6
	Ours	96.2	82.0
Ours	Ours	**96.7**	**83.0**

* means using top$k$ pruning scheme in [5].

TABLE III
QUANTITATIVE COMPARISON ON THE ETH DATASET. RESULTS WITH DIFFERENT PAIRWISE REGISTRATION ALGORITHMS ARE REPORTED, INCLUDING FCGF [26] AND YOHO [28]

Pose graph	Method	FCGF [26] RR(%)	YOHO [28] RR(%)
Full	EIGSE3 [41]	44.8	60.9
	L1-IRLS [53]	60.5	77.2
	RotAvg [53]	67.3	85.4
	LITS [47]	26.3	34.8
	HARA [54]	72.2	85.4
	SGHR [5]	85.7	98.8
	Ours	96.8	98.2
Pruned	EIGSE3 [41]	89.4	96.3
	L1-IRLS [53]	86.1	90.2
	RotAvg [53]	95.6	96.6
	LITS [47]	41.2	48.4
	HARA [54]	90.3	96.0
	SGHR [5]*	97.4	97.2
	SGHR [5]	96.8	99.1
	Ours	97.0	99.4
Ours	Ours	**97.9**	**99.6**

* means using top$k$ pruning scheme in [5].

3DLoMatch. The best performance is achieved on the ScanNet dataset, where the proposed method obtained a 2.8° reduction in the mean rotation error and a 0.1 m reduction in the mean translation error under pruning condition. ETH is an outdoor point cloud dataset in which the proposed method achieves 0.5% improvement in RR for YOHO and 1.1% for FCGF. The result of the ETH dataset demonstrates the generalization ability of our approach to outdoor scenarios. The positive performance on three different types of datasets demonstrates the good performance of the proposed method. Especially, further dilation of the performance gap on datasets other than 3DMatch reveals that our method is not bothered by the biased nature of learning-based approaches.

We also notice that when canceling the application of the pruning scheme and constructing the fully connected graph directly, SGHR and our method simultaneously show some degree of performance degradation. However, the proposed method still maintains a relatively good performance when constructing fully connected graphs. Especially on the ScanNet dataset, the rotation error and translation error of SGHR on the fully connected graph without pruning drop a lot compared

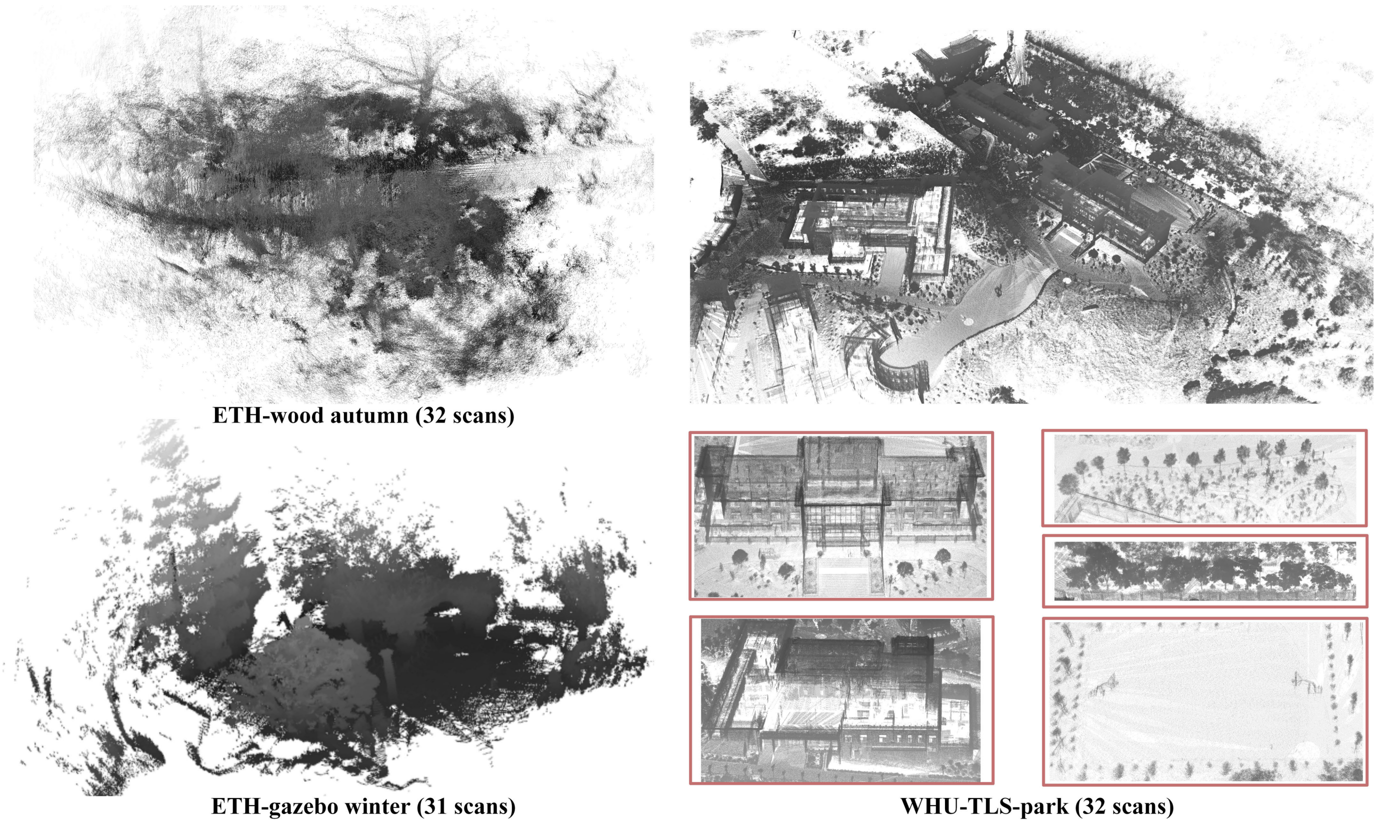

Fig. 4. Qualitative result on (left) ETH and (right) WHU-TLS datasets, featuring outdoor scenes like wood summer, gazebo winter, and park. The red boxes highlight the specific details of the park scene in WHU-TLS.

to that on the sparse graph with 4.7° in mean rotation error and 0.14 m in mean translation error. However, the proposed method only drops 1.2° in rotation error and 0.1 m in translation error. The rotation error gap between our method and SGHR on ScanNet increased from 2.8° on the pruned sparse pose graph to 6.3° on the fully connected graph, proving our method has better robustness to fully connected pose graphs with more outliers. Simultaneously, after switching to the proposed pruning strategy, the performance of our method is further improved, which verifies the effectiveness of the proposed pruning strategy.

E. Analysis

1) Registration Results in Low Overlap Scenarios: To further analyze the performance of the proposed method in low-overlapping scenarios, we filter the park scenarios in the WHU-TLS dataset with an overlap rate lower than 0.4. The results are illustrated in Fig. 5. For low overlap point cloud registration, one of the major challenges is to find overlap regions. However, our score estimation method obviously has better robustness in low overlap. It can be seen that in the initial pose graph, the correct and incorrect edges are assigned high and low weights, respectively. In addition, as the iterations progress, the weights of these edges return to where they should be. This indicates that the proposed method is capable of accurately discriminating correct edges from incorrect edges and continuously reducing the weight of outliers during the iteration process to minimize their impact on the global pose. The reason is that during the weight update stage, we not only consider the influence of the current error but also guide the weight update based on historical information.

2) Correlation Between the Predicted Overlap Score and the GT Overlap Ratio: A well-calibrated overlap score could filter scan pairs with greater overlap regions, indicating that scan pairs are easily aligned by pairwise registration. We report the relationship between the predicted overlap score and the GT overlap ratio on the 3DMatch dataset and the ETH dataset. As shown in Fig. 6, the horizontal coordinate of each point represents the ground-truth (GT) overlap rate, and the vertical coordinate represents the predicted overlap score. It can be seen that there is a positive correlation between the true overlap rate and the predicted overlap score. This exhibits that pairs of point clouds with high overlap rates are given larger overlap scores by the proposed method. It is also noticeable that the points are approximately distributed along both sides of the line $y = x$, which fully demonstrates the accuracy of the predicted overlap scores.

3) GT Overlap Ratios With Top-k Overlap Scores: We also report the average GT overlap ratio for top-k scan pair results. Fig. 7 shows that the average GT overlap ratio continues to decrease as k increases, indicating that the postadded point cloud pairs have a smaller GT overlap ratio. This implies that our overlap score can accurately select point cloud pairs with a high GT overlap ratio. This is because the proposed overlapping score estimation method uses the algebraic connectivity of the graph to predict overlapping regions and estimate

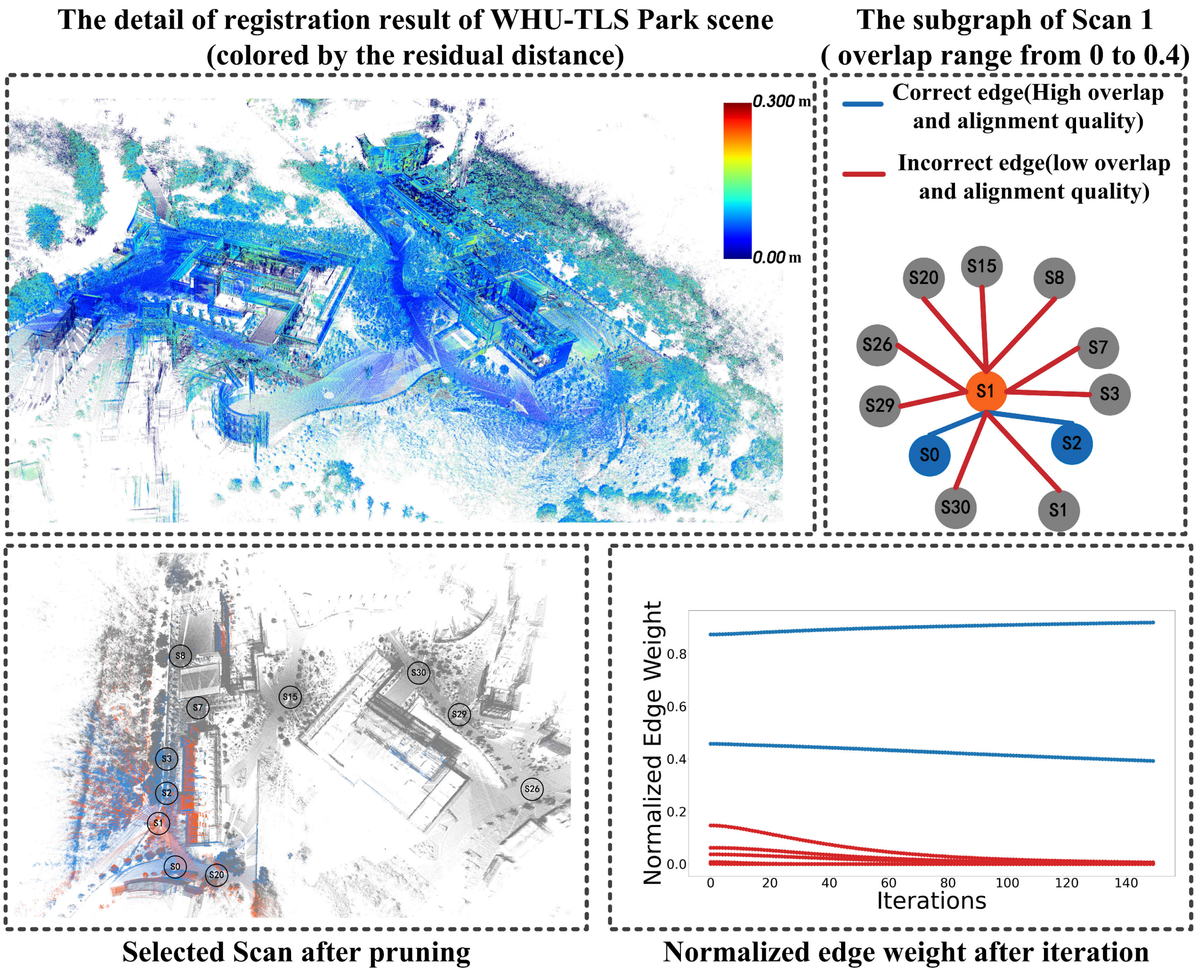

Fig. 5. Detail result of the proposed method in park scenarios (32 scans) from the WHU-TLS dataset, with an overlap rate range from 0 to 0.4. The left of the first row is the registration result colored by residual distance, and the right shows the selected ten scans after pruning with scan 1 as an example. The left of the second row displays the positional distribution of these ten scans, and the right visualizes the change of weights after iteration.

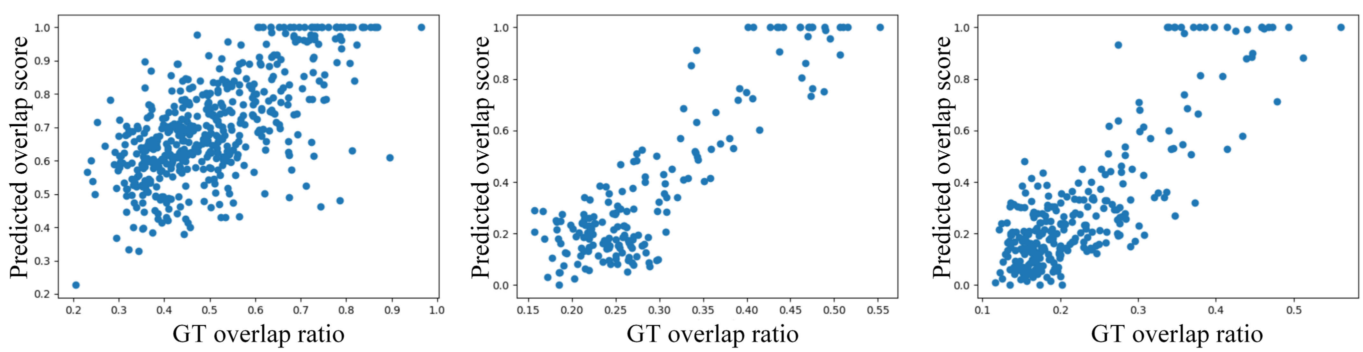

Fig. 6. Relationship between the predicted overlap score and the GT overlap ratio on the 3DMatch dataset and the ETH dataset, (left) featuring 3DMatch-kitchen, (center) ETH-gazebo summer, and (right) ETH-gazebo winter. The predicted overlap score is normalized to the range [0, 1].

overlapping degree scores, which has better robustness than directly comparing feature similarity and reduces the influence of global blurring. By retaining highly overlapping point cloud pairs through a pruning scheme, our method can diminish the impact of outliers and obtain a good initial pose graph.

F. Ablation Study

Our ablation studies of the proposed method include two main parts: the construction of the pose graph and the proposed history reweighting function in the IRLS algorithm. The results are in Table IV.

1) Pose Graph Construction: We predict confidence using the SD of the generated spatial compatibility matrix. In this experiment, alternatively, we generate the initial pose graph for testing by employing the weight estimation approaches in SGHR [5], LMVR [6], and UOR [8], respectively. By comparing the confidence levels of the local correspondence, our method can assign more reliable weights to the initial pose graph without bias to the scene. The results show that

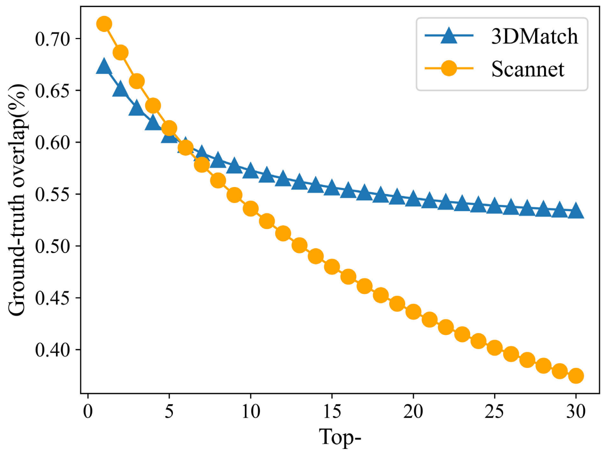

Fig. 7. Averaged GT overlap ratios with top-k overlap scores.

TABLE IV

ABLATION STUDY ON THE 3DMATCH AND ETH DATASETS

	3DMatch	ETH	ScanNet	
	RR(%)	RR(%)	RE(°)	TE(m)
NetVLAD [5]+HWA	89.5	97.2	24.3	0.55
LMVR [6]+HWA	95.5	99.3	19.8	0.55
UOR [8]+HWA	96.1	99.2	19.7	0.54
SD+HR [5]	96.3	99.1	19.4	0.58
SD+HWA(ours)	**96.7**	**99.6**	**18.9**	**0.44**

our confidence prediction method based on SD has better reliability on all three datasets.

2) Reweighting Function: Motivated by [5], we design a reweighting function based on the HWA. To validate the effectiveness of the proposed reweighting function, we use the history reweighting function (HR) in [5] as an alternative, combined with the proposed confidence method (SD). Compared with HR, HWA exhibits a notable improvement, which demonstrates that our reweighting function is more robust to outliers.

3) Sensitivity of Parameters: To achieve optimal parameter selection and conduct sensitivity analysis, we provide results for two crucial parameters θ_d and K across various settings on 3DMatch. θ_d is a distance parameter to control the sensitivity to variations in length. An excessively large K tends to increase the computational cost and include outlier edges whereas too small K could divide the entire graph into multiple disconnected subgraphs. Fig. 8 shows that our algorithm achieves the best performance at $\theta_d = 0.11$ or 0.15, and $K = 10$ on 3DMatch.

4) Robustness for Different Overlap Ratios and Point Cloud Densities: Limited overlap and density pose significant challenges to the registration of the point cloud. To further validate the robustness and generalization capabilities of the proposed method, we provide additional analyzes by altering different overlap ratios and density levels. In this context, we consider three settings for overlap and density. The results are shown in Tables V and VI. It can be seen that our algorithm performs better in "high overlap" compared to "low overlap." One possible explanation is that, in scenarios with

TABLE V

ABLATION STUDY ON VARYING OVERLAP RATIOS. "LOW OVERLAP," "MID OVERLAP," AND "HIGH OVERLAP" CORRESPOND TO OVERLAP RATIOS IN THE RANGE OF 0%–30%, 30%–60%, AND 60%–100% FOR 3DMATCH AND SCANNET AND 0%–20%, 20%–40%, AND 40%–60% FOR ETH

	3DMatch	ETH	ScanNet	
	RR(%)	RR(%)	RE(°)	TE(m)
Low overlap	83.0	97.8	23.2	0.54
Mid overlap	96.6	100	6.2	0.14
High overlap	98.5	100	1.8	0.02

TABLE VI

ABLATION STUDY ON 3DMATCH WITH VARYING DENSITY. DIFFERENT DENSITIES ARE MANIFESTED AS DIFFERENT NUMBERS OF RANDOMLY SAMPLED KEY POINTS

Points Number	1500	3000	5000
	RR(%)	RR(%)	RR(%)
3DMatch	93.7	95.0	96.7

TABLE VII

DETAILED TIME CONSUMPTION FOR REGISTERING THE SCENES ON WHU-TLS. "GRAPH CONS" MEANS THE TIME FOR CONSTRUCTING THE INPUT POSE GRAPH. "TRANS SYNC" MEANS THE TIME FOR IRLS-BASED TRANSFORMATION SYNCHRONIZATION

Name	Scans	Pts (million)	Graph Cons (s)	Trans Sync (s)	Total (s)
Park	32	160.24	15.1	1.7	16.8
Campus	10	109.05	4.7	0.4	5.1
Mountain	6	19.61	3.4	0.2	3.6
River bank	13	93.11	3.8	0.2	4.0
Tunnel	7	157.02	3.7	0.2	3.9
Excavation	12	482.42	5.0	0.5	5.5

TABLE VIII

COMPARISON OF TIME CONSUMPTION ON THE SCANNET DATASET

Name	Graph Cons (s)	Trans Sync (s)	Total (s)
SGHR	7.2	2.0	9.2
Ours	11.6	1.8	13.4

low overlap, the correspondences tend to fall into nonoverlapping regions, which leads to an increased rate of outliers. This observation also motivated us to develop a pruning scheme based on the overlap ratio. Similar results could be found at different density levels. Notably, the commendable results in scenarios with limited overlap and density illustrate the strong robustness and generalization capabilities of our method.

G. Running Time

To verify the efficiency of our method in processing large-scale point cloud datasets, we tested the proposed method on the WHU-TLS dataset. Table VII reports the detailed time consumption of our method for different scenarios in the WHU-TLS dataset. The findings indicate that the number of scans exerts a more significant influence on the time consumption than the number of points because more scans will result in more nodes in the graph. Holistically, the

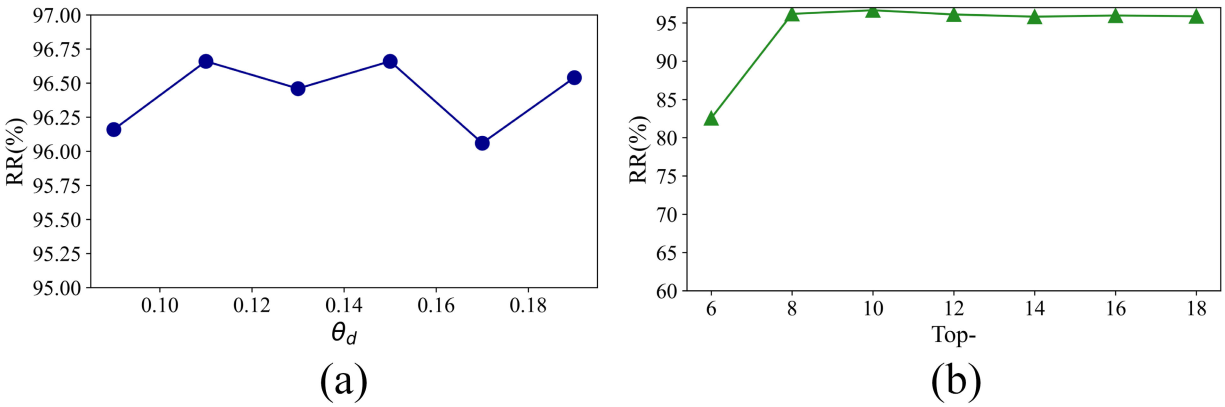

Fig. 8. Sensitivity of parameters θ_d and K tested on 3DMatch. (a) and (b) blue circles and green triangles represent the RR performance under different parameter settings, respectively.

proposed method still maintains high efficiency in aligning large-scale point cloud datasets.

SGHR represents the state-of-the-art methodology, demonstrating superior performance in both accuracy and computational efficiency. To substantiate the efficiency of our method, we conducted a comparative analysis with SGHR, focusing on time consumption. Table VIII reveals that our method exhibits higher graph construction time consumption relative to SGHR. SGHR constructs the pose graph by extracting global features via a trained network, exhibiting a speed superior to that of the SD approach. However, reliance on extensive training data will introduce biases inherent in training scenes. The proposed method obviates the need for training data, thus avoiding these potential biases and extensive data annotation work.

V. Conclusion

In this article, we have introduced a multiview point cloud registration method based on SD. Different from previous works that predict the point cloud overlap rate as the confidence for pairwise transformation matrices, our approach focuses on evaluating the quality of the corresponding set which is used to estimate the relative pose. This evaluation naturally serves as a prediction of the confidence of the pairwise transformation matrix. We perform this evaluation by obtaining the confidence score for each corresponding set, which is computed by the SD of the constructed spatial compatibility matrices. In addition, to improve computational efficiency and mitigate interference at outlier edges, we propose a pruning scheme based on the algebraic connectivity of the graph. Finally, we present a reweighting function based on the HWA and an error calculation based on the exponential moving average, which is robust to outliers. We conduct exhaustive experiments on three datasets: 3D(Lo)Match, ScanNet, and ETH. The experimental results demonstrate that our method has good performance and generalization on both indoor and outdoor datasets.

The limitations of the proposed method are primarily twofold. Initially, when there are two point clouds with low overlap that cannot form a loop through other point clouds, the proposed method is unable to construct a completed pose graph. Additionally, spectral techniques are weak in time complexity. Consequently, applying the proposed method to real-time tasks remains inadequate. Future research could concentrate on enhancing efficiency to enable applications in real-time tasks.

References

[1] F. Pomerleau, F. Colas, and R. Siegwart, "A review of point cloud registration algorithms for mobile robotics," *Found. Trends Robot.*, vol. 4, no. 1, pp. 1–104, 2015.

[2] P. Kim, J. Chen, and Y. K. Cho, "SLAM-driven robotic mapping and registration of 3D point clouds," *Automat. Construct.*, vol. 89, pp. 38–48, May 2018.

[3] Y. Zheng, Y. Li, S. Yang, and H. Lu, "Global-PBNet: A novel point cloud registration for autonomous driving," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 11, pp. 22312–22319, Nov. 2022.

[4] M. Sinko, P. Kamencay, R. Hudec, and M. Benco, "3D registration of the point cloud data using ICP algorithm in medical image analysis," in *Proc. ELEKTRO*, May 2018, pp. 1–6.

[5] H. Wang et al., "Robust multiview point cloud registration with reliable pose graph initialization and history reweighting," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 9506–9515.

[6] Z. Gojcic, C. Zhou, J. D. Wegner, L. J. Guibas, and T. Birdal, "Learning multiview 3D point cloud registration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 1759–1769.

[7] X. Huang, Z. Liang, X. Zhou, Y. Xie, L. J. Guibas, and Q. Huang, "Learning transformation synchronization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 8074–8083.

[8] J. Zhao, Q. Zhu, Y. Wang, W. Peng, H. Zhang, and J. Mao, "Registration of multiview point clouds with unknown overlap," *IEEE Trans. Multimedia*, vol. 3, pp. 1–16, 2023.

[9] Z. Dong, B. Yang, F. Liang, R. Huang, and S. Scherer, "Hierarchical registration of unordered TLS point clouds based on binary shape context descriptor," *ISPRS J. Photogramm. Remote Sens.*, vol. 144, pp. 61–79, Oct. 2018.

[10] B. Yang, Z. Dong, F. Liang, and Y. Liu, "Automatic registration of large-scale urban scene point clouds based on semantic feature points," *ISPRS J. Photogramm. Remote Sens.*, vol. 113, pp. 43–58, Mar. 2016.

[11] D. F. Huber and M. Hebert, "Fully automatic registration of multiple 3D data sets," *Image Vis. Comput.*, vol. 21, no. 7, pp. 637–650, Jul. 2003.

[12] X. Ge, Q. Zhu, L. Huang, S. Li, and S. Li, "Global registration of multiview unordered forest point clouds guided by common subgraphs," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022, Art. no. 5700214, doi: 10.1109/TGRS.2021.3057643.

[13] A. S. Mian, M. Bennamoun, and R. Owens, "Three-dimensional model-based object recognition and segmentation in cluttered scenes," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 28, no. 10, pp. 1584–1601, Oct. 2006.

[14] M. Leordeanu and M. Hebert, "A spectral technique for correspondence problems using pairwise constraints," in *Proc. 10th IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2005, pp. 1482–1489.

[15] X. Bai et al., "PointDSC: Robust point cloud registration using deep spatial consistency," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 15859–15869.

[16] Z. Chen, K. Sun, F. Yang, and W. Tao, "SC2-PCR: A second order spatial compatibility for efficient and robust point cloud registration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 13221–13231.

[17] J. Lee, S. Kim, M. Cho, and J. Park, "Deep Hough voting for robust global registration," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 15994–16003.

[18] S. Quan and J. Yang, "Compatibility-guided sampling consensus for 3-D point cloud registration," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7380–7392, Oct. 2020.

[19] J. Yang, Z. Huang, S. Quan, Z. Qi, and Y. Zhang, "SAC-COT: Sample consensus by sampling compatibility triangles in graphs for 3-D point cloud registration," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022, Art. no. 5700115, doi: 10.1109/TGRS.2021.3058552.

[20] A. Zeng, S. Song, M. Nießner, M. Fisher, J. Xiao, and T. Funkhouser, "3DMatch: Learning local geometric descriptors from RGB-D reconstructions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 199–208.

[21] S. Huang, Z. Gojcic, M. Usvyatsov, A. Wieser, and K. Schindler, "PREDATOR: Registration of 3D point clouds with low overlap," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 4265–4274.

[22] A. Dai, A. L. S. Chang, M. Savva, M. Halber, T. Funkhouser, and M. Nießner, "ScanNet: richly-annotated 3D reconstructions of indoor scenes," in *IEEE Proc. Comput. Vis. Pattern Recognit. (CVPR)*, Jan. 2017, pp. 1–26.

[23] F. Pomerleau, M. Liu, F. Colas, and R. Siegwart, "Challenging data sets for point cloud registration algorithms," *Int. J. Robot. Res.*, vol. 31, no. 14, pp. 1705–1711, Dec. 2012.

[24] X. Bai, Z. Luo, L. Zhou, H. Fu, L. Quan, and C. Tai, "D3Feat: Joint learning of dense detection and description of 3D local features," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 6358–6366.

[25] S. Ao, Q. Hu, B. Yang, A. Markham, and Y. Guo, "SpinNet: Learning a general surface descriptor for 3D point cloud registration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 11753–11762.

[26] C. Choy, J. Park, and V. Koltun, "Fully convolutional geometric features," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 8957–8965.

[27] Z. Gojcic, C. Zhou, J. D. Wegner, and A. Wieser, "The perfect match: 3D point cloud matching with smoothed densities," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5540–5549.

[28] H. Wang, Y. Liu, Z. Dong, and W. Wang, "You only hypothesize once: Point cloud registration with rotation-equivariant descriptors," in *Proc. 30th ACM Int. Conf. Multimedia*, Oct. 2022, pp. 1630–1641.

[29] H. Wang et al., "RoReg: Pairwise point cloud registration with oriented descriptors and local rotations," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 8, pp. 10376–10393, Aug. 2023.

[30] H. Yang and L. Carlone, "A polynomial-time solution for robust registration with extreme outlier rates," in *Robotics: Science and Systems (RSS)*, 2019. [Online]. Available: http://rss2019.informatik.uni-freiburg.de/papers/0013_FI.pdf

[31] H. M. Le, T.-T. Do, T. Hoang, and N.-M. Cheung, "SDRSAC: semidefinite-based randomized approach for robust point cloud registration without correspondences," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 124–133.

[32] P. J. Besl and N. D. McKay, "Method for registration of 3-D shapes," in *Sensor Fusion IV: Control Paradigms and Data Structures*, vol. 1611. Bellingham, WA, USA: SPIE, 1992, pp. 586–606.

[33] S. Rusinkiewicz and M. Levoy, "Efficient variants of the ICP algorithm," in *Proc. 3rd Int. Conf. 3-D Digit. Imag. Modeling*, May 2001, pp. 145–152.

[34] S. D. Billings et al., "Iterative most-likely point registration (IMLP): A robust algorithm for computing optimal shape alignment," *PLoS ONE*, vol. 10, pp. 1–45, Mar. 2015.

[35] X. Huang, J. Zhang, L. Fan, Q. Wu, and C. Yuan, "A systematic approach for cross-source point cloud registration by preserving macro and micro structures," *IEEE Trans. Image Process.*, vol. 26, no. 7, pp. 3261–3276, Jul. 2017.

[36] K. Fu, S. Liu, X. Luo, and M. Wang, "Robust point cloud registration framework based on deep graph matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 8893–8902.

[37] W. Yuan, B. Eckart, K. Kim, V. Jampani, D. Fox, and J. Kautz, "DeepGMR: Learning latent Gaussian mixture models for registration," in *Computer Vision—ECCV*. Cham, Switzerland: Springer, 2020, pp. 733–750.

[38] Q. Li, R. Xiong, and T. Vidal-Calleja, "A GMM based uncertainty model for point clouds registration," *Robot. Auto. Syst.*, vol. 91, pp. 349–362, May 2017.

[39] P. W. Theiler, J. D. Wegner, and K. Schindler, "Globally consistent registration of terrestrial laser scans via graph optimization," *ISPRS J. Photogramm. Remote Sens.*, vol. 109, pp. 126–138, Nov. 2015.

[40] T. Birdal, U. Simsekli, M. O. Eken, and S. Ilic, "Bayesian pose graph optimization via Bingham distributions and tempered geodesic MCMC," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2018, pp. 1–20.

[41] F. Arrigoni, B. Rossi, and A. Fusiello, "Spectral synchronization of multiple views in SE(3)," *SIAM J. Imag. Sci.*, vol. 9, no. 4, pp. 1963–1990, Jan. 2016, doi: 10.1137/16M1060248.

[42] U. Bhattacharya and V. M. Govindu, "Efficient and robust registration on the 3D special Euclidean group," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 5884–5893.

[43] S. Choi, Q.-Y. Zhou, and V. Koltun, "Robust reconstruction of indoor scenes," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 5556–5565.

[44] P. W. Holland and R. E. Welsch, "Robust regression using iteratively reweighted least-squares," *Commun. Statist., Theory Methods*, vol. 6, no. 9, pp. 813–827, Jan. 1977.

[45] F. Bernard, J. Thunberg, P. Gemmar, F. Hertel, A. Husch, and J. Goncalves, "A solution for multi-alignment by transformation synchronisation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 2161–2169.

[46] X. Huang, Z. Liang, C. Bajaj, and Q. Huang, "Translation synchronization via truncated least squares," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, Dec. 2017, pp. 1459–1468.

[47] Z. J. Yew and G. H. Lee, "Learning iterative robust transformation synchronization," in *Proc. Int. Conf. 3D Vis. (3DV)*, Dec. 2021, pp. 1206–1215.

[48] M. Fiedler, "Laplacian of graphs and algebraic connectivity," *Banach Center Publications*, vol. 25, no. 1, pp. 57–70, 1989.

[49] A. Jamakovic and S. Uhlig, "On the relationship between the algebraic connectivity and graph's robustness to node and link failures," in *Next Generation Internet Networks*. Piscataway, NJ, USA: IEEE Press, 2007, pp. 96–102.

[50] A. Jamakovic and P. V. Mieghem, "On the robustness of complex networks by using the algebraic connectivity," in *Proc. Int. Conf. Res. Netw.*, Jan. 2008, pp. 183–194.

[51] M. Arie-Nachimson, S. Z. Kovalsky, I. Kemelmacher-Shlizerman, A. Singer, and R. Basri, "Global motion estimation from point matches," in *Proc. 2nd Int. Conf. 3D Imag., Modeling, Process., Vis. Transmiss.*, Oct. 2012, pp. 81–88.

[52] Z. Dong et al., "Registration of large-scale terrestrial laser scanner point clouds: A review and benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 163, pp. 327–342, May 2020.

[53] A. Chatterjee and V. M. Govindu, "Robust relative rotation averaging," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 958–972, Apr. 2018.

[54] S. H. Lee and J. Civera, "HARA: A hierarchical approach for robust rotation averaging," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 15777–15786.

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